# Machine Learning 2026 • Tutorial 2

Estimating $\theta_1$ and $\theta_2$: Maximum Likelihood, MAP, and the Full Posterior **SOLUTIONS**

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Today's distribution is one you have not met, and its parameters are called $\theta_1$ and $\theta_2$ — nothing else. That is deliberate: in this course a model's unknowns are always $\boldsymbol{\theta}$, whether the model is this curve, a logistic regression, or a network with a million weights. Nothing here can be recalled; it has to be derived. **No calculator is needed.** Work in the boxes.

## Part 1 The model, and the two unknowns

You have trained a model. Now you are using it: you hand it one input, it thinks, and it gives you back an answer. That takes time — call it $X$, measured with a stopwatch, in milliseconds.

Do it eight times and you do not get the same number eight times. Most inputs are easy and come back quickly; the occasional awkward one takes far longer. So $X$ is never negative, most of its values bunch up small, and a few stretch a long way to the right. The standard model for a quantity shaped like that is the **lognormal**: $X$ is lognormal when $\log X$ is normal. Writing $\theta_1$ for the centre and $\theta_2$ for the variance of $\log X$,

$$p(x \mid \theta_1, \theta_2) \;=\; \frac{1}{x\sqrt{2\pi\theta_2}}\,\exp\!\left(-\frac{(\log x - \theta_1)^2}{2\theta_2}\right), \qquad x > 0.$$

Here are the eight times, in milliseconds. Because only $\log x$ appears above, both rows are given:

| time to answer $x_i$ (ms) | 7.4 | 20.1 | 20.1 | 54.6 | 54.6 | 54.6 | 148.4 | 1096.6 |
|---|---|---|---|---|---|---|---|---|
| $\log x_i$ | 2 | 3 | 3 | 4 | 4 | 4 | 5 | 7 |

log **means the natural logarithm, base** $e \approx 2.718$ **— as it does everywhere in this course, and as** `np.log` **means it in NumPy.** The eight times were chosen as $e^2, e^3, \ldots, e^7$ so that the second row is whole numbers: $e^2 = 7.39$, $e^3 = 20.09$, $e^4 = 54.60$, $e^5 = 148.41$, $e^7 = 1096.63$. If you reach for the log key on a calculator you will get base 10 and none of this will work.

**(a)** Look at the two rows. Why is a model for $\log X$ a more sensible thing to fit here than a model for $X$ itself?

**Answer.** The times span two orders of magnitude and are badly skewed — the slowest answer takes 150 times as long as the fastest. Their logs span 2 to 7 and sit far more evenly. A symmetric bell is a poor description of the first row and a reasonable one of the second, and it is the second that the model actually uses.

## Part 2 Steps 1–3: one observation, all $n$, then logs

**(a)** Write $L(\theta_1, \theta_2)$ for the eight times, then $\ell = \log L$. Write $y_i = \log x_i$ to keep it short, and separate the terms that contain the unknowns from those that do not.

**Answer.**

$$L = \prod_{i=1}^{n} \frac{1}{x_i \sqrt{2\pi\theta_2}} \exp\!\left(-\frac{(y_i - \theta_1)^2}{2\theta_2}\right),$$

$$\ell = \underbrace{-\sum_i \log x_i - \tfrac{n}{2}\log 2\pi}_{\text{no unknowns}} - \frac{n}{2}\log\theta_2 - \frac{1}{2\theta_2}\sum_{i=1}^{n}(y_i - \theta_1)^2.$$

The first group is a constant: it shifts $\ell$ up or down but cannot move where the maximum is, so it can be ignored from here on.

## Part 3 Step 4 for $\theta_1$

**(a)** Differentiate $\ell$ with respect to $\theta_1$, set it to zero, and solve. Then evaluate.

**Answer.** Only the last term contains $\theta_1$:

$$\frac{\partial \ell}{\partial \theta_1} = \frac{1}{\theta_2}\sum_{i=1}^{n}(y_i - \theta_1) = 0 \implies \sum_i y_i - n\theta_1 = 0 \implies \widehat{\theta}_1 = \frac{1}{n}\sum_{i=1}^{n} y_i.$$

The logs sum to $2+3+3+4+4+4+5+7 = 32$, so $\widehat{\theta}_1 = 32/8 = \mathbf{4}$. Note $\theta_2$ cancelled: the centre does not depend on the spread.

## Part 4 Step 4 for $\theta_2$

**(a)** Now differentiate with respect to $\theta_2$, set to zero, and solve. Then evaluate, using $S = \sum_i (y_i - \widehat{\theta}_1)^2$.

**Answer.**

$$\frac{\partial \ell}{\partial \theta_2} = -\frac{n}{2\theta_2} + \frac{1}{2\theta_2^2} \sum_i (y_i - \theta_1)^2 = 0 \implies -n\theta_2 + S = 0 \implies \widehat{\theta}_2 = \frac{S}{n}.$$

Deviations from 4: $-2, -1, -1, 0, 0, 0, 1, 3$. Squares: $4, 1, 1, 0, 0, 0, 1, 9$, so $S = 16$ and $\widehat{\theta}_2 = 16/8 = \mathbf{2}$.

**(b)** Both parts of Step 4 used the same three moves. Name them.

**Answer.** Differentiate with respect to the unknown; set the derivative to zero; solve. The only thing that changed between $\theta_1$ and $\theta_2$ was which symbol was held fixed. That is the whole method — it does not care which parameter, or which distribution.

## Part 5 Turning the prior into a posterior

Hold $\theta_1$ at 4. Only $\theta_2$ is unknown now.

The problem now hands you one extra piece of information. You are not the first person to time this model. Whoever had it before you left a note:

> *"We timed it as well. The spread of* $\log X$ *came out near* 1 *— but we only ran it a handful of times."*

So two answers are on the table, and they disagree. The note says the spread is about 1. Your eight numbers say 2. Which do you believe?

Neither, on its own. The note is called a **prior**, and this part combines it with your eight numbers to get one answer, called the **posterior**.

To combine them, the prior has to be written as a curve over $\theta_2$. Since $\theta_2$ is a spread it can never be negative, and the usual curve for a quantity like that is the **inverse-gamma**. It carries two numbers, $a$ and $b$:

$$p(\theta_2) \;\propto\; \theta_2^{-a-1} \exp\!\left(-\frac{b}{\theta_2}\right), \qquad \text{here } a = 3 \text{ and } b = 4.$$

Think of $a$ as how much weight the belief carries, and $b$ as the total those imagined observations bring with them. This prior carries weight $a + 1 = 4$. Today's data will carry weight $\frac{n}{2} = 4$ — for a spread, each timing counts a half. So the two sides of this contest enter *exactly tied*, on purpose: you will see the tie in the answer.[1]

**This whole part in one line.** Prior IG$(3, 4)$, times eight numbers, gives posterior IG$(7, 12)$ — and you get there by adding exponents, not by integrating anything.

Here is why it is that easy. The likelihood from Part 4 and the prior above are the same shape:

$$\underbrace{\theta_2^{-n/2}\, e^{-S/2\theta_2}}_{\text{likelihood}} \qquad \text{and} \qquad \underbrace{\theta_2^{-a-1}\, e^{-b/\theta_2}}_{\text{prior}}$$

Both are *(a power of* $\theta_2$*)* $\times$ exp$(-a$ *constant*$/\theta_2)$. Multiply two things of that shape and the powers add and the constants add. Nothing else can happen.

**(a)** Fill the table. The prior row is done for you; use $n = 8$ and $S = 16$, and put in numbers, not symbols.

| | **power of** $\theta_2$ | **constant inside the** exp |
|---|---|---|
| prior, $a = 3,\ b = 4$ | $-a - 1 = -4$ | $b = 4$ |
| likelihood, $n = 8,\ S = 16$ | $-n/2 = -4$ | $S/2 = 8$ |
| **product** = posterior | **−8** | **12** |

**(b)** An inverse-gamma$(a, b)$ has power $-a-1$ and constant $b$. Read your product row against that and name the posterior.

**Answer.** The product is $\theta_2^{-8} e^{-12/\theta_2}$. Matching, $-a - 1 = -8$ so $a = 7$, and $b = 12$: the posterior is **inverse-gamma**$(7, 12)$ — the same family the prior came from. A prior that comes back in its own family after meeting the data is called **conjugate**. In symbols, what the table just did in numbers is

$$p(\theta_2 \mid x) \;\propto\; \theta_2^{-(a+n/2)-1} \exp\!\left(-\frac{b + S/2}{\theta_2}\right), \qquad a \to a + \tfrac{n}{2} = 7, \quad b \to b + \tfrac{S}{2} = 12.$$

[1] Most books write these two numbers $\alpha$ and $\beta$; they are the same two numbers. The full density carries a constant $b^a/\Gamma(a)$ in front, where $\Gamma$ is the factorial extended to numbers that are not whole. You will never evaluate it here — everything below is done up to proportionality, so it cancels.

**(c)** Your posterior is IG$(7, 12)$. The mode of an inverse-gamma$(a, b)$ is $b/(a+1)$ and its mean is $b/(a-1)$. Give the MAP estimate and the posterior mean, and say why they are not the same number.

**Answer.** $\widehat{\theta}_2^{\,\mathrm{MAP}} = 12/8 = \mathbf{1.5}$; posterior mean $= 12/6 = \mathbf{2.0}$. They differ because the posterior is skewed: the mode is where the curve is highest, the mean is where it balances, and for a right-skewed curve the balance point sits to the right of the peak.

And there is the tie you were promised. The prior alone says $\theta_2 = 1.0$ (its mode); the data alone say 2.0 (the MLE); the MAP is 1.5 — *exactly halfway*, because weight 4 met weight 4:

$$\widehat{\theta}_2^{\,\mathrm{MAP}} = \frac{b + \frac{S}{2}}{(a+1) + \frac{n}{2}} = \frac{4 \cdot 1.0 + 4 \cdot 2.0}{4 + 4} = 1.5.$$

The MAP is a weighted average of what the prior says and what the data say. That is not special to this problem — it is what these updates always are.

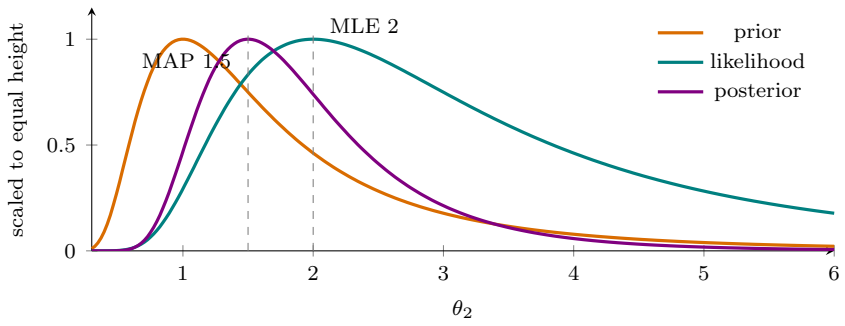
**(d)** You are holding a whole distribution, not a number: its central 95% runs from about 0.9 to 4.3. So which one number should the model use — MLE 2.0, MAP 1.5, or posterior mean 2.0?

**Answer.** First, what the interval says that no single number could: eight observations do not pin $\theta_2$ down. All three estimates sit inside a range running from below 1 to above 4, so the data are consistent with half, or twice, that spread.

**Where it can, a good model uses none of the three.** After only eight times the parameter genuinely *is* a distribution, and the honest prediction for how long the next answer takes averages over all of it:

$$p(x_{\mathrm{new}} \mid \mathrm{data}) = \int p(x_{\mathrm{new}} \mid \theta_2)\, p(\theta_2 \mid \mathrm{data})\, \mathrm{d}\theta_2.$$

When one number is unavoidable, the choice follows from what it is for: the **posterior mean** under squared-error cost, and it is the usual default; the **MAP** if you want the single most probable value, which is what regularised training returns; the **MLE** only if the prior holds nothing worth keeping. With an interval this wide, *which* centre you pick matters far less than reporting the width at all.




*Prior, likelihood and posterior, each scaled to the same height so their positions can be compared. The posterior sits between the other two and is narrower than either.*

**Where we end today — two things to take with you, neither of which you have to derive.**

**1. This pattern has a name, and you have seen it before.** A prior that comes back in its own family after meeting the data is called **conjugate**, and in every case the prior's parameters are counts and totals to which the data simply adds its own:

| Data model | Unknown | Prior | Posterior |
|---|---|---|---|
| Bernoulli | a proportion | Beta$(a, b)$ | Beta$(a + k,\ b + n - k)$ |
| Normal, spread known | the centre | Normal$(m, s^2)$ | Normal, both updated |
| **Normal, centre known** | **the spread** | **Inv-gamma**$(a, b)$ | **Inv-gamma**$\left(a + \frac{n}{2},\ b + \frac{S}{2}\right)$ |
| Poisson | a rate | Gamma$(a, b)$ | Gamma$(a + \sum x_i,\ b + n)$ |

Lecture 3's Beta prior was the first row. Today was the third.

**2. MAP is regularisation.** Take logs of the MAP objective and flip the sign:

$$\underbrace{\arg\max_{\boldsymbol{\theta}} \big[\log p(x \mid \boldsymbol{\theta}) + \log p(\boldsymbol{\theta})\big]}_{\text{MAP}} = \arg\min_{\boldsymbol{\theta}} \big[\underbrace{-\log p(x \mid \boldsymbol{\theta})}_{\text{loss}} + \underbrace{-\log p(\boldsymbol{\theta})}_{\text{penalty}}\big].$$

Fitting with no penalty is the MLE — least squares is exactly this. Fitting with one is the MAP, and when the model is a regression the penalty has the names you will meet later: a Gaussian prior on the weights gives ridge, a Laplace prior gives lasso, and the prior's strength is $\lambda$. The prior here carried weight 4, exactly what eight timings carry, which is why the answer landed exactly halfway between what the prior said and what the data said — and why, on eight hundred timings, the prior would hardly move it at all.

***Later in this course:*** *This is the difference between fitting a model and knowing how much to trust it.*

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The four steps used today are the four from Lecture 3: write $p(x_i \mid \boldsymbol{\theta})$, multiply over the $n$ observations, take logs, then maximise — by calculus where that works, and by argument where it does not. **After the tutorial:** worked solutions are released on the portal, with the two profile curves and the prior–likelihood–posterior picture drawn out. The recipe itself is always available at `ml2026.floodresq.com/explore/mle`.